

Question 2: Elevator Simulation

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Algorithm

1. Start
2. Input the number of requests, N .
3. While $N < 1$, ask again.
4. Set CurrentFloor $\leftarrow 0$ and Count $\leftarrow 1$.
5. While Count $\leq N$:
 1. Input Requested Floor.
 2. If RequestedFloor $>$ CurrentFloor, display "Moving up".
 3. Otherwise, if RequestedFloor $<$ CurrentFloor, display "Moving down".
 4. Otherwise, display "doors opening".
 5. Set ~~the~~ CurrentFloor $\leftarrow$ RequestedFloor.
 6. Increase Count by 1.
6. STOP.

Pseudo-code:-

Begin

Input be N .

While ($N < 1$), do.

print "Enter at least one request".

Input N

END While.

CurrentFloor $\leftarrow 0$

Count $\leftarrow 1$

While (Count $\leq N$), do

Input Requested Floor

```
IF (RequestedFloor > CurrentFloor), then  
    print "Moving Up".  
ELSE IF (RequestedFloor < CurrentFloor), then  
    print "Moving down"  
ELSE  
    print "Doors opening".  
END IF
```

```
CurrentFloor  $\leftarrow$  RequestedFloor.  
Count  $\leftarrow$  Count + 1
```

```
END While
```

```
STOP.
```

